CELLS

Module map

- Module 04: Cells
- Connected lab: lab-04_liquid-chemistry.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 04

Cells

1. Cell theory
2. Prokaryotic and eukaryotic organization
3. Membranes and compartments
4. Organelles and division of labor
5. Microscopy and evidence

Lab evidence

lab-04_liquid-chemistry.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

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Learning objectives

- State the core claims of cell theory.
- Compare prokaryotic and eukaryotic cells using structure and scale.
- Explain how membranes and compartments support cell function.
- Match major organelles to their roles.
- Use microscope evidence to support claims about cells.

OBJECTIVE 1
State the core claims of cell theory.

OBJECTIVE 2
Compare prokaryotic and eukaryotic cells using structure and scale.

OBJECTIVE 3
Explain how membranes and compartments support cell function.

OBJECTIVE 4
Match major organelles to their roles.

OBJECTIVE 5
Use microscope evidence to support claims about cells.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

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Topic sequence

- Cell theory: Cell theory explains cells as the basic units of life and as descendants of existing cells.
- Prokaryotic and eukaryotic organization: Prokaryotic and eukaryotic cells differ in compartmentalization and internal organization.
- Membranes and compartments: Membranes separate internal conditions from the environment and define organelles.
- Organelles and division of labor: Organelles divide cellular work such as energy capture, protein processing, and storage.
- Microscopy and evidence: Microscopy turns cell structure into observable evidence.

01
Cell theory
Cell theory explains cells as the basic units of life and as descendants of existing cells.

02
Prokaryotic and eukaryotic organization
Prokaryotic and eukaryotic cells differ in compartmentalization and internal organization.

03
Membranes and compartments
Membranes separate internal conditions from the environment and define organelles.

04
Organelles and division of labor
Organelles divide cellular work such as energy capture, protein processing, and storage.

05
Microscopy and evidence
Microscopy turns cell structure into observable evidence.

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

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Module 04: Cells Evidence Map

- Module focus: Cells
- Central claim: Cells explains cell theory with compartmental structure and microscopy evidence.
- Key nodes to connect: Cell theory, Prokaryotic and eukaryotic organization, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-04_liquid-chemistry.md supplies an evidence surface for the map.

Teaching note: Ask students to explain one edge in their own words before moving on.

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Module 04: Cells Reasoning Sequence

- Module focus: Cells
- Trace the model: Cell theory -> Prokaryotic and eukaryotic organization -> Membranes and compartments -> Organelles and division of labor
- Inputs become outputs: Cell theory, Prokaryotic and eukaryotic organization -> State the core claims of cell theory., Compare prokaryotic and eukaryotic cells using structure and scale.
- Feedback check: lab-04_liquid-chemistry.md evidence checks whether students can explain membranes and compartments.
- Lab connection: lab-04_liquid-chemistry.md gives students a process to observe or test.

Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

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Terms as evidence handles

- Cell theory: Principle that organisms are made of cells and cells come from cells.
- Prokaryote: Cell without a membrane-bound nucleus.
- Eukaryote: Cell with a membrane-bound nucleus and organelles.
- Organelle: Specialized cell structure with a particular job.
- Nucleus: Eukaryotic compartment that stores DNA.
- Mitochondrion: Organelle central to aerobic energy harvesting.

CELL THEORY
Principle that organisms are made of cells and cells come from cells.

PROKARYOTE
Cell without a membrane-bound nucleus.

EUKARYOTE
Cell with a membrane-bound nucleus and organelles.

ORGANELLE
Specialized cell structure with a particular job.

NUCLEUS
Eukaryotic compartment that stores DNA.

MITOCHONDRION
Organelle central to aerobic energy harvesting.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

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Lab connection

- Lab file: lab-04_liquid-chemistry.md
- Lab evidence should test or illustrate: Membranes and compartments
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Cell theory

EVIDENCE
lab-04_liquid-chemistry.md

CLAIM
State the core claims of cell theory.

Teaching note: Make the lab connection explicit before students encounter the activity.

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Contrast check

- Do not stop at: Cell theory explains cells as the basic units of life and as descendants of existing cells.
- Stronger explanation: Microscopy turns cell structure into observable evidence.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE
Cell theory explains cells as the basic units of life and as descendants of existing cells.

DEEPER
Microscopy turns cell structure into observable evidence.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

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Module 04: Cells Retrieval and Lab Check

- Module focus: Cells
- Retrieval prompt: What does cell theory claim?
- Answer check: State the core claims of cell theory.
- Required terms: Cell theory, Prokaryote, Eukaryote
- Lab connection: lab-04_liquid-chemistry.md; lab-04_liquid-chemistry.md supplies evidence for water, pH, and solution conditions that shape cellular behavior.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

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Practice quiz bridge

- 1. Which statement is part of cell theory?
- 2. What is the clearest structural difference between a prokaryotic and a eukaryotic cell?
- 3. A cell is packed with mitochondria. This most likely means the cell:
- 4. Which structure would you find in a plant cell but not an animal cell?

QUESTION 1

Key: B

Which statement is part of cell theory?

QUESTION 2

Key: D

What is the clearest structural difference between a prokaryotic and a eukaryotic cell?

QUESTION 3

Key: A

A cell is packed with mitochondria. This most likely means the cell:

QUESTION 4

Key: C

Which structure would you find in a plant cell but not an animal cell?

Teaching note: Use the quiz as formative feedback. Answer key: 1=B, 2=D, 3=A, 4=C

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Synthesis and exit ticket

- Exit claim: explain cell theory using evidence from lab-04_liquid-chemistry.md.
- Must include: Cell theory, Prokaryote, Eukaryote.
- Revision prompt: what evidence would change your explanation?

CLAIM
Cell theory

EVIDENCE
lab-04_liquid-chemistry.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.